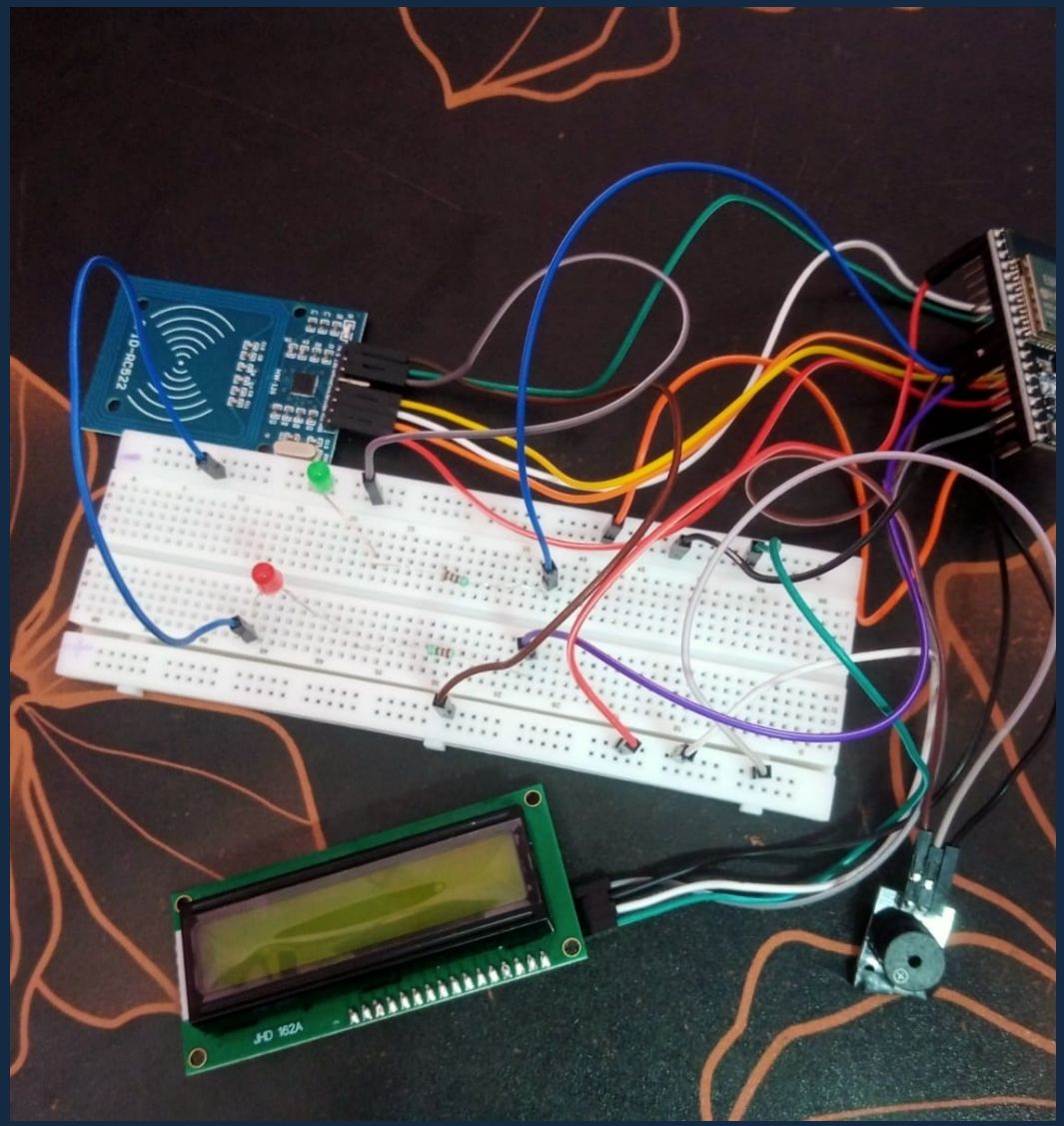
IOT & ROBOTICS | MAJOR PROJECT

SMART ATTENDANCE SYSTEM USING RFID

ESP32 • RC522 RFID • 16x2 I2C LCD
LED and buzzer feedback

PROJECT DOCUMENTATION
Source code, circuit and test results



Prototype photograph supplied by project team

Problem Statement & Aim

Why an RFID-based approach is useful

Problem

Manual attendance can take time and can lead to recording errors. Maintaining consistent and easily retrievable attendance records can be difficult with purely manual methods.

Project aim

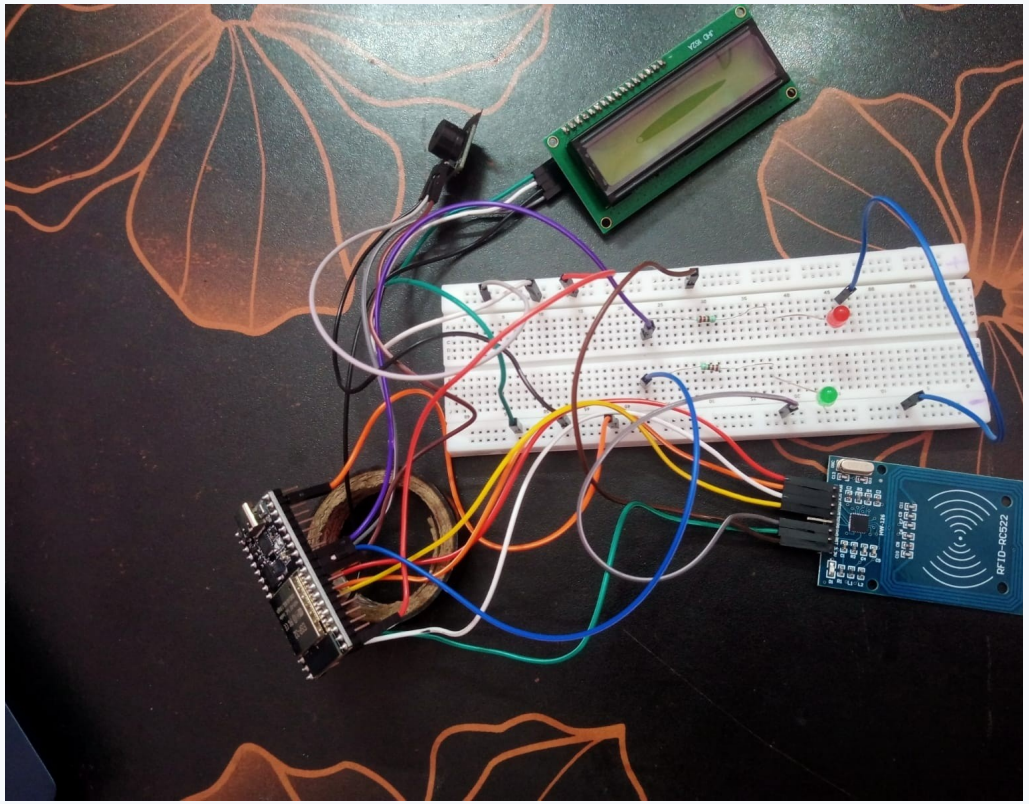
Build a small prototype that reads an RFID card, checks whether its UID is registered, and provides immediate LCD, LED, buzzer, and Serial Monitor feedback.

RFID helps automate card identification; a card UID alone does not prove the cardholder's identity.

Objectives

Functions implemented in the supplied firmware

- Read RC522 cards through SPI on the ESP32.
- Map registered card UIDs to student names.
- Store student registrations in ESP32 Preferences.
- Show prompts and status messages on a 16x2 LCD.
- Use red/green LEDs and a buzzer for feedback.
- Prevent a second attendance mark for the same UID during the current power session.



Physical assembly documented in the supplied photo.

Hardware Components

Components used in the completed and tested prototype

ESP32

Controller and program storage.

RC522 RFID

Reads a compatible card/tag UID.

16x2 I2C LCD

Local prompts and result messages.

Green + red LEDs

Visual feedback through 220 Ω resistors.

Buzzer module

Provides audible feedback for scan results.

Breadboard + wires

Prototyping interconnections.

System Architecture

Data and control flow



Persistent data

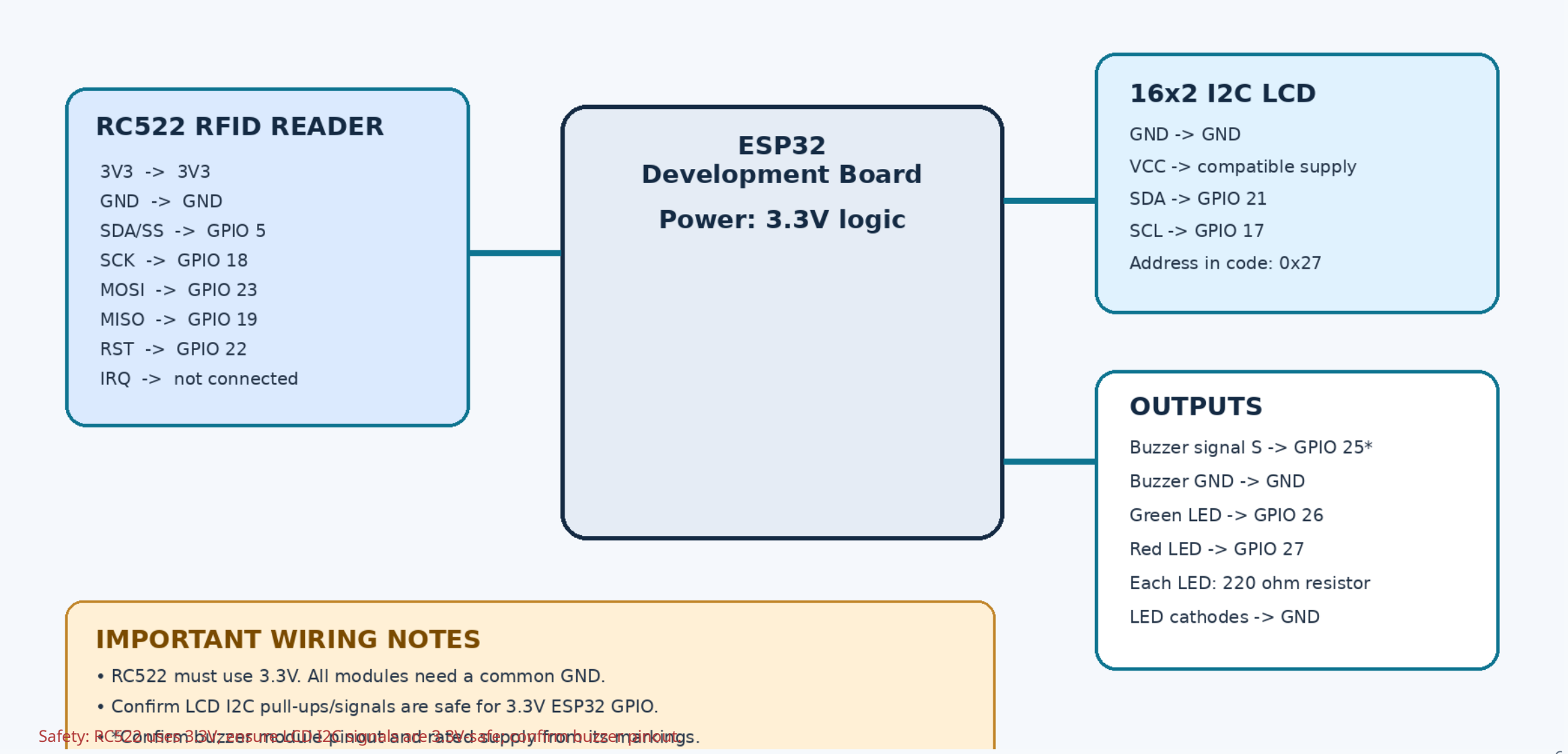
Registered UID/name mappings stored using ESP32 Preferences.

Session data

Attendance de-duplication is in RAM and resets after power cycle.

Circuit Design & Pin Mapping

Pin mapping used in the tested prototype



Working Principle

Card scan decision flow

1. Startup

Initialize Serial, GPIO, LCD, SPI and RC522.

2. Load records

Read registered student UID/name pairs from Preferences.

3. Scan card

Read and normalize the RFID UID.

4. Check UID

Unknown UID -> red LED and not-registered message.

5. Check duplicate

Already marked this session -> already-marked message.

6. Mark attendance

Registered, not yet marked -> green LED, name message and success beep.

Software Features & Serial Commands

Open Serial Monitor at 115200 baud; select Newline as line ending

Command	Purpose
HELP	Shows command list and usage.
LIST	Lists registered cards/names.
ADD <UID> <Name>	Registers or updates a card UID/name.

Example format

ADD 04:A1:B2:C3 Student One

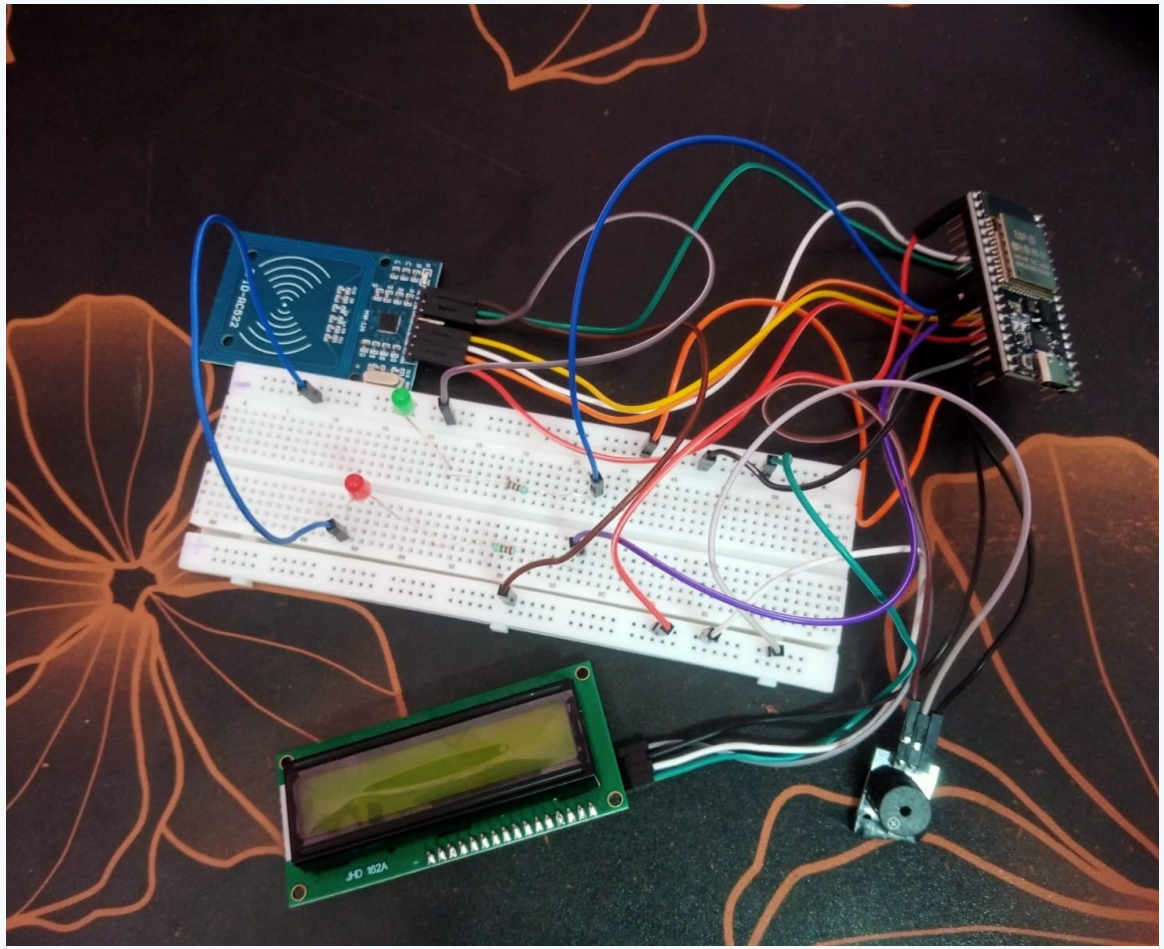
Replace the example UID with the exact UID printed after a real card scan.

Time limitation

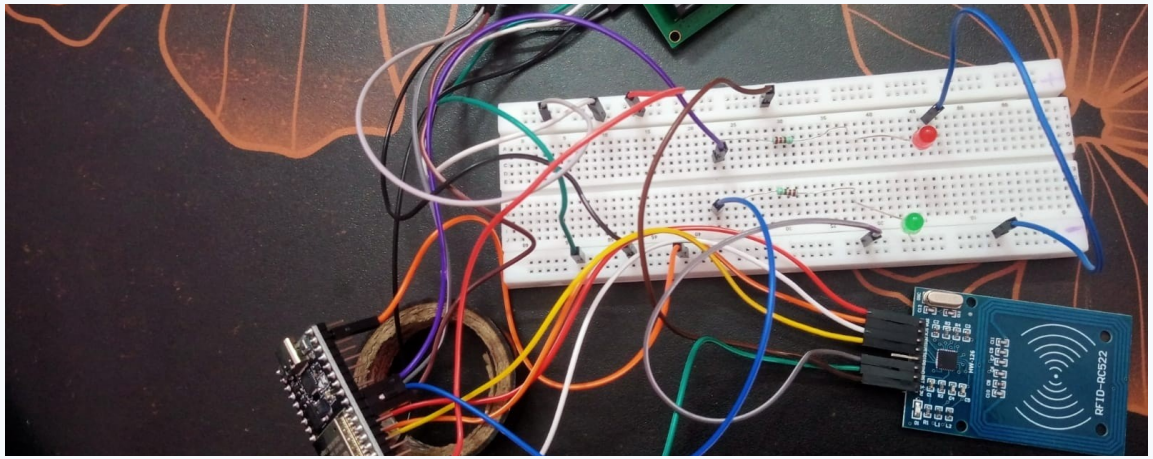
The code logs elapsed uptime (millis), not a calendar date/time. An RTC or synchronized time source would be needed for real timestamps.

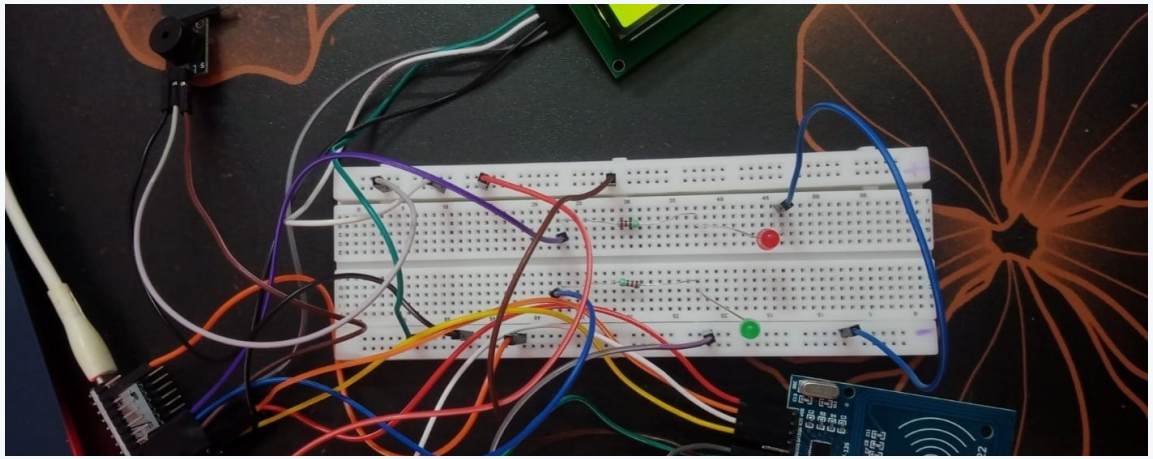
Prototype Photos

The pictures document the physical assembly



View 1





Alternate views

Testing & Results

All functional tests completed successfully on the assembled prototype

Assembly

Completed prototype assembled and documented in photographs.

Compilation & upload

Firmware compiled and uploaded to the ESP32 successfully.

LCD display

Startup message and “Scan your card” prompt displayed correctly.

RFID card tests passed

Registered card: attendance marked and student name displayed. Unregistered card: “Not registered”. Repeat scan: “Already marked”; no duplicate attendance during the current session.

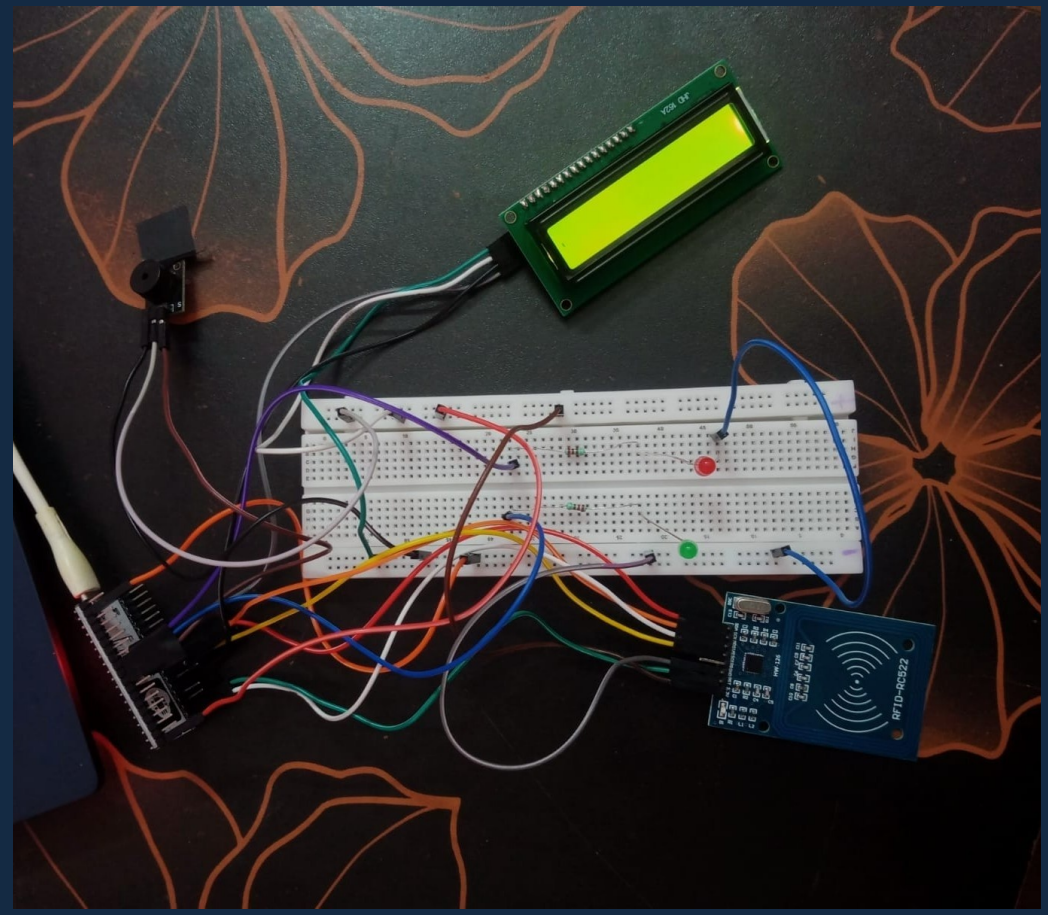
Indicators & Serial output

Green LED and success beep confirmed accepted cards; red LED and error beeps flagged unregistered cards. Serial Monitor displayed card UIDs and attendance events.

CONCLUSION

The RFID attendance workflow was implemented and tested successfully.

The source code supports UID registration, student-name lookup, LCD status messages, LED/buzzer feedback, session duplicate checking, and serial logging.



Result: all functional tests in the implemented project scope passed successfully.